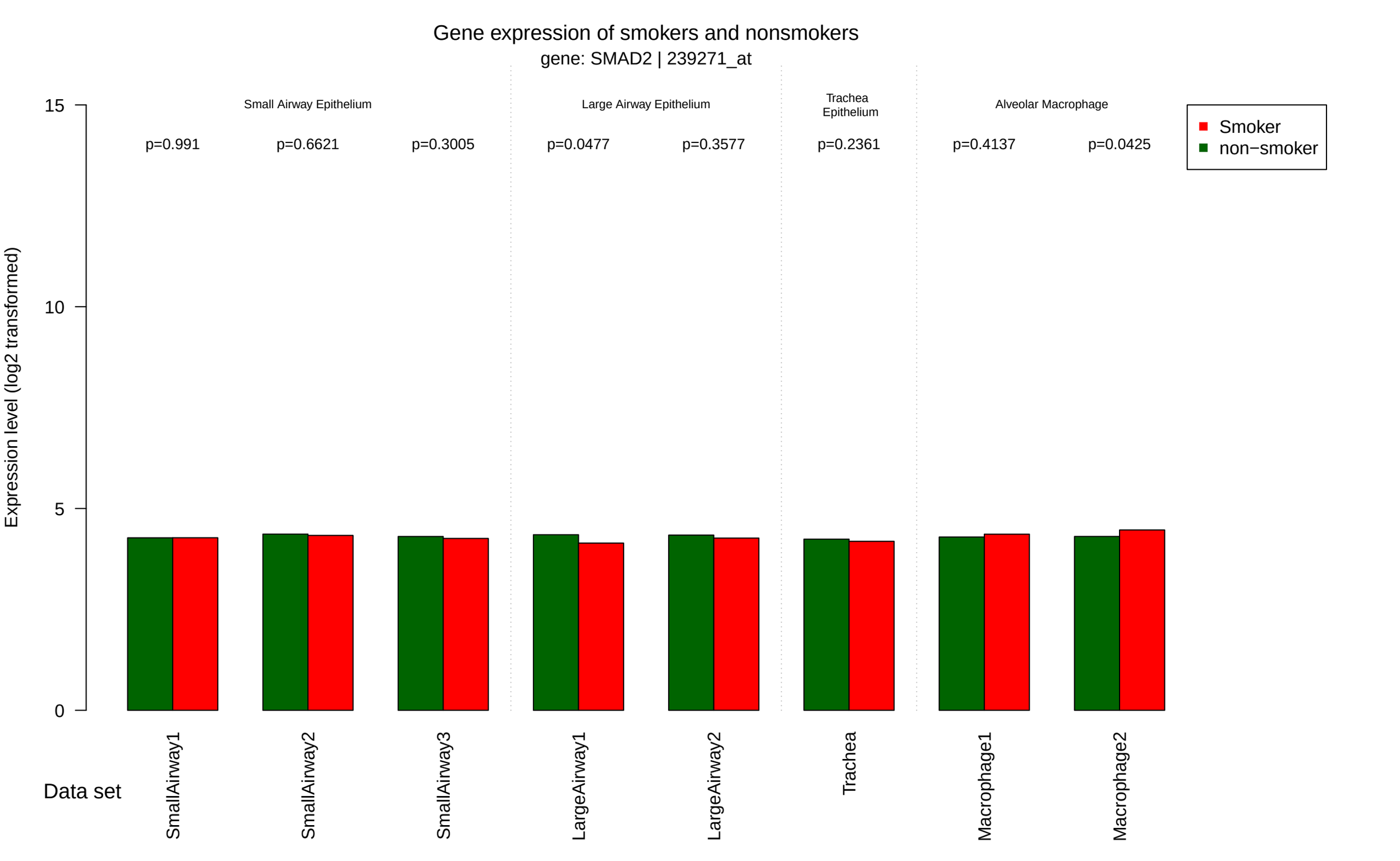
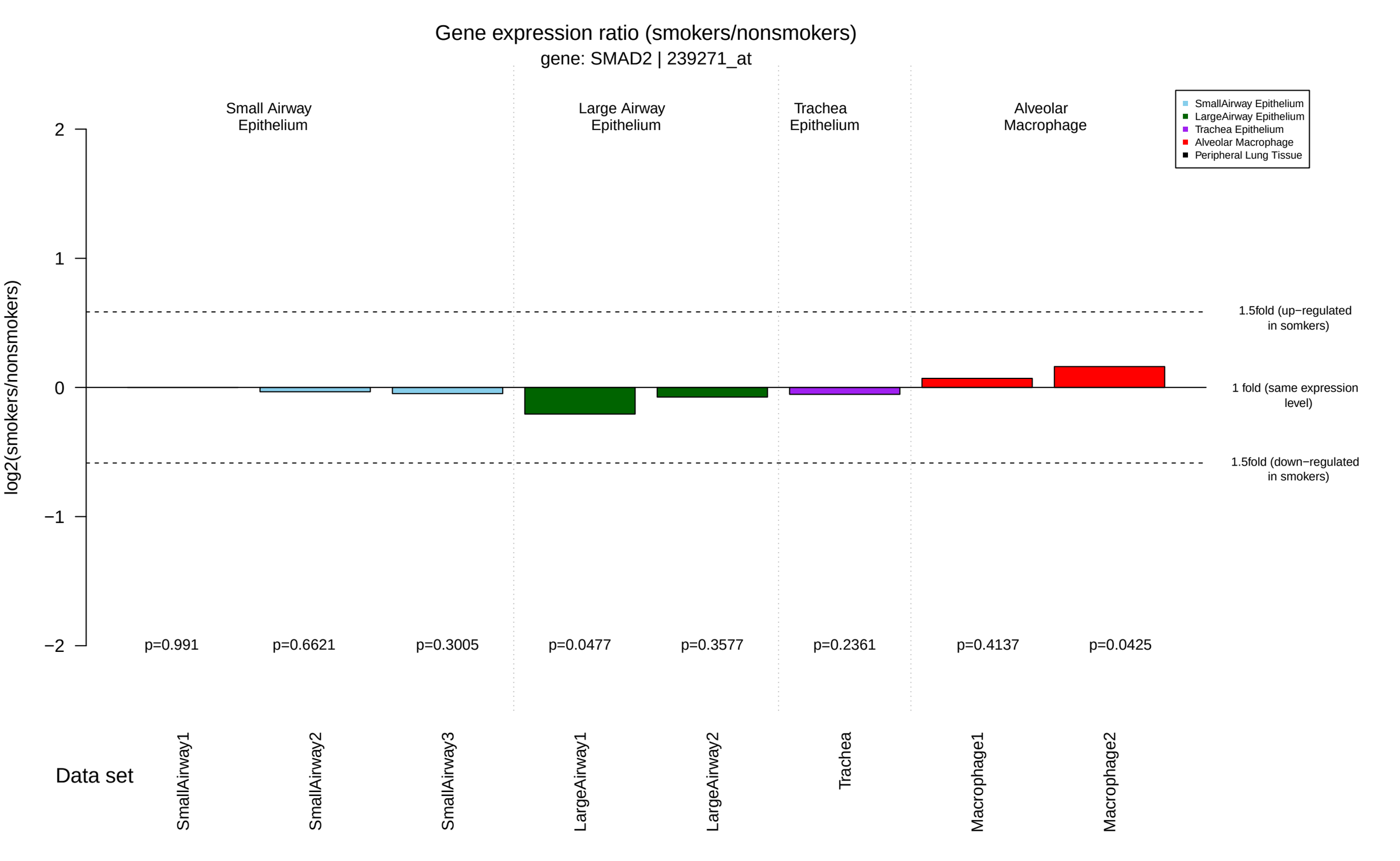
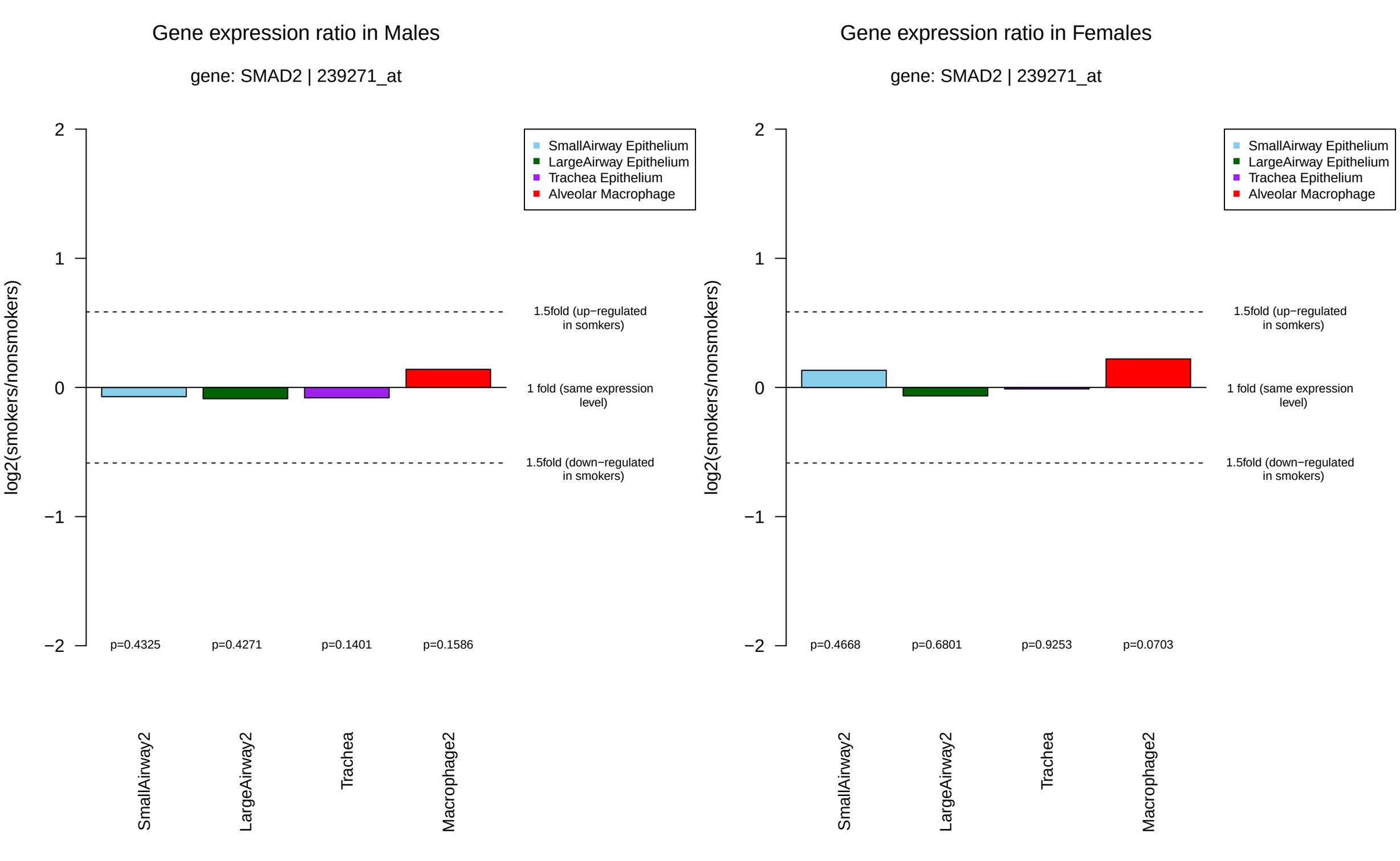
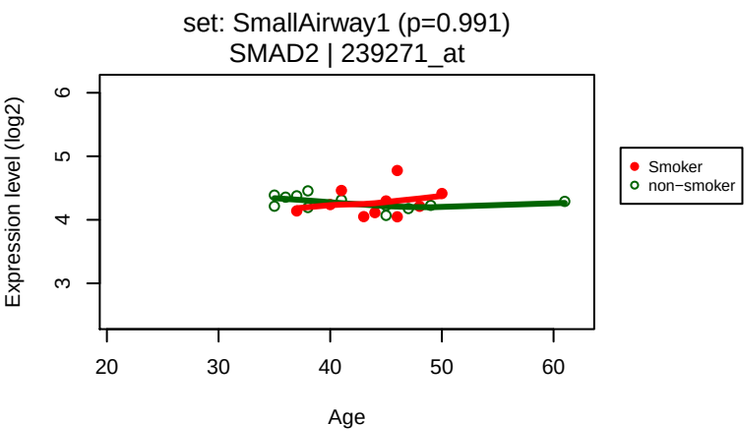
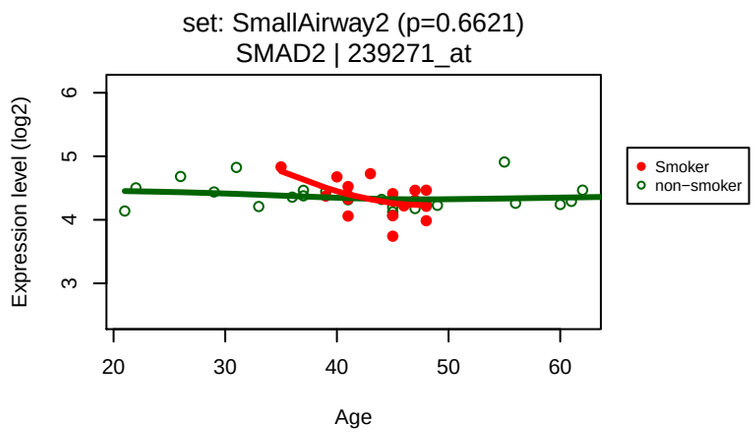
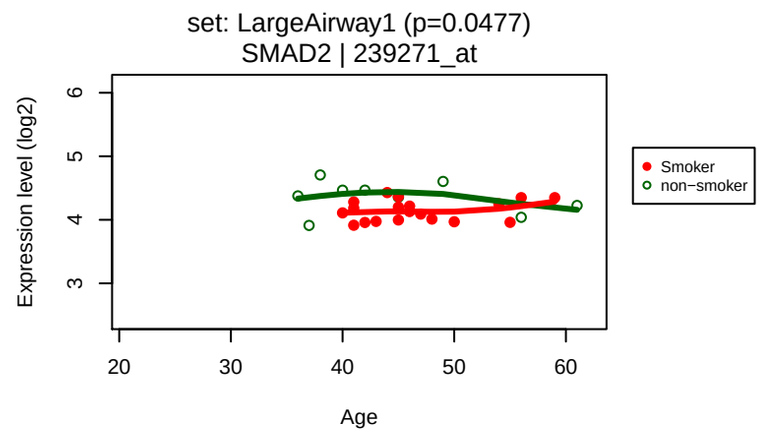

Gene expression of smokers and nonsmokers
gene: SMAD2 | 239271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.991
p=0.6621
p=0.3005
p=0.0477
p=0.3577
p=0.2361
p=0.4137
p=0.0425
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

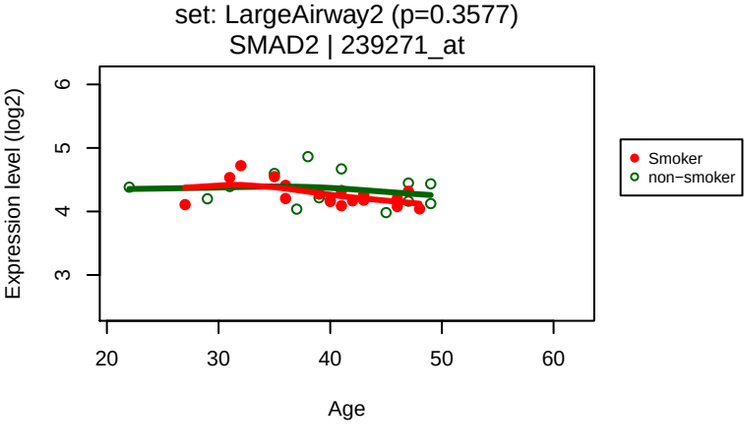

Gene expression ratio (smokers/nonsmokers)
gene: SMAD2 | 239271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.991
p=0.6621
p=0.3005
p=0.0477
p=0.3577
p=0.2361
p=0.4137
p=0.0425
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

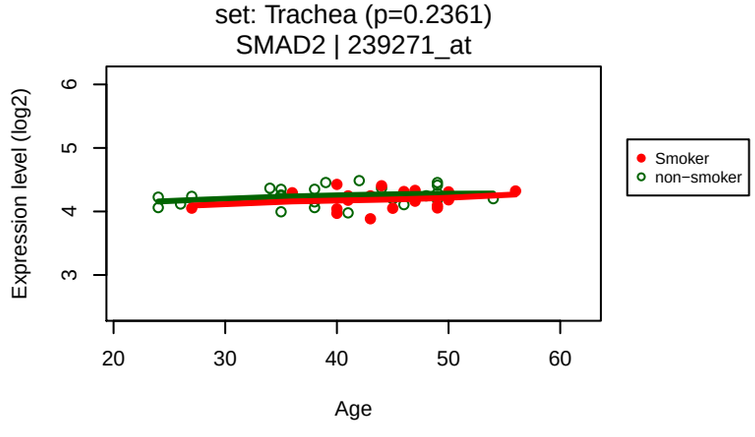

Gene expression ratio in Males
gene: SMAD2 | 239271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4325
p=0.4271
p=0.1401
p=0.1586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD2 | 239271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4668
p=0.6801
p=0.9253
p=0.0703
SmallAirway2
LargeAirway2
Trachea
Macrophage2

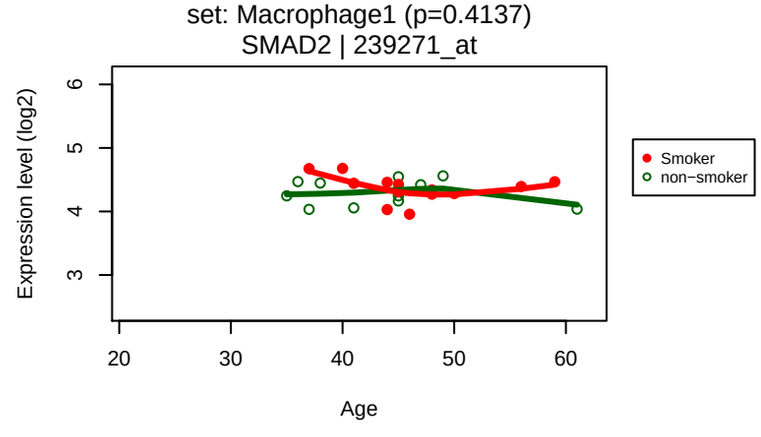

set: SmallAirway1 (p=0.991)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker

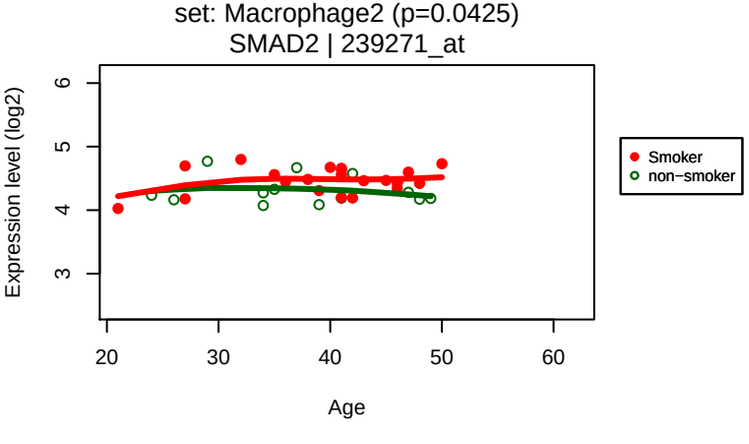

set: SmallAirway2 (p=0.6621)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker

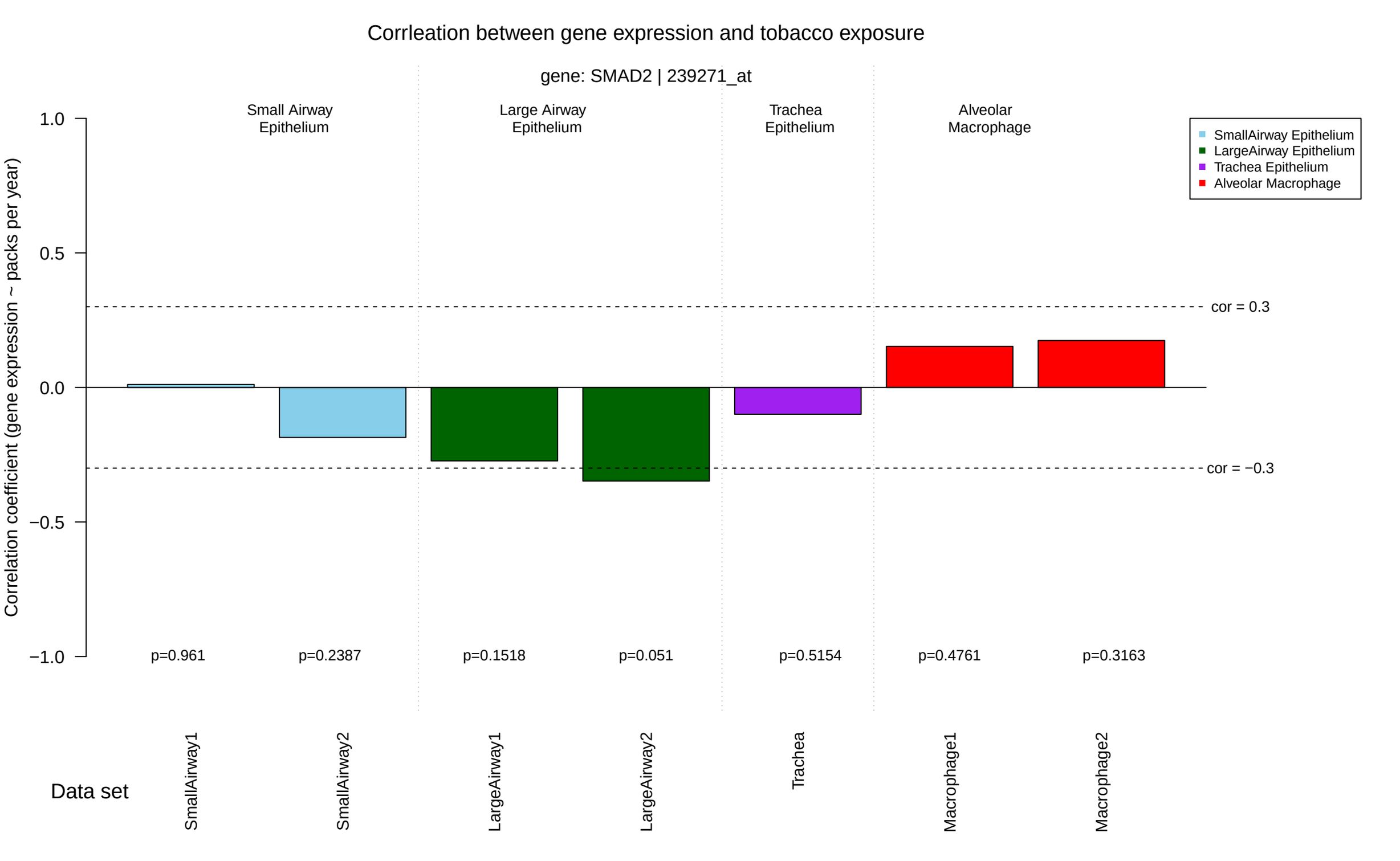

set: LargeAirway1 (p=0.0477)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3577)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2361)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4137)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0425)
SMAD2 | 239271_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD2 | 239271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.961
p=0.2387
p=0.1518
p=0.051
p=0.5154
p=0.4761
p=0.3163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2